PCCE E-CELL HACKATHON · 2026

HeatWatch 🌡️

Science-Grade Urban Heat Island Prediction for City Planners

Real-time satellite intelligence that turns NASA data into decisions city planners can act on — in 60 seconds.

Team RAMpage

THE PROBLEM

Cities Are Flying Blind on Heat

> Mumbai's urban core runs **4–6°C hotter** than its outskirts. India recorded **700+ heat-related deaths** in 2024 alone.

🏛️ Planners Lack Precision

District-level weather data can't pinpoint which neighborhood is a heat emergency. Block-by-block intelligence simply doesn't exist.

📡 Tools Are Inaccessible

NASA and ISRO satellite data exists — but requires GIS software, Python, and weeks of processing. No city official has time for that.

🌳 No Decision Support

Even when data exists, planners can't simulate "what if we plant trees here?" — so budgets get spent on guesswork, not outcomes.

Sources: IMD Heat Action Reports 2024 · WHO Urban Heat Health Risk Framework

THE SOLUTION

HeatWatch: From Satellite to Decision in 60 Seconds

We transform raw NASA satellite imagery into actionable intelligence — no GIS expertise required.

Real MODIS Satellite Data

Live LST + NDVI tiles from NASA via Google Earth Engine — actual satellite imagery overlaid on your city map, not estimates.

ONNX ML Risk Engine

4-pillar UHI scoring: temperature delta, vegetation loss, urban density, and land surface temp — rural vs. urban baseline in real time.

Intervention Simulator

Add 20% tree cover to a neighborhood — sliders instantly show projected °C reduction and budget estimate. Decisions backed by data.

Community Heat Reporting

Citizens drop pins to report unbearable heat or lack of shade — ground truth no satellite can provide, transforming HeatWatch into a civic tool.

MARKET OPPORTUNITY

A Massive and Mandated Market

$27.4B

TAM

Global Climate Tech SaaS market by 2030

$3.2B

SAM

Urban climate intelligence platforms for municipalities and agencies

$180M

SOM

India Smart City + urban heat resilience mandates across 4,800+ ULBs

The Mandate Gap

India's National Action Plan on Climate Change requires Heat Action Plans for all Tier 1 and Tier 2 cities.

None of them have a real-time UHI monitoring tool.

HeatWatch is built to fill that gap — starting now.

Sources: MarketsandMarkets 2024 · MoHUA Smart Cities Mission · NDMA Heat Action Plan 2023

BUSINESS MODEL

B2B SaaS + Community Freemium

Tier	Who	Price	What They Get
Free	Citizens, researchers	₹0	Global location search, Real-time weather data, Heat zone detection
Pro	Urban planners, NGOs	₹4,999/month	ML-powered risk assessment intervention simulator,
Enterprise	Municipalities, state govts	Custom	Custom integrations, Dedicated support team, bulk city analysis

TECH STACK

Built on Open Data. Infinitely Scalable.

🛰️ Data & AI Layer

- NASA MODIS (MOD11A1 LST · MOD13A1 NDVI) via Google Earth Engine
- OpenWeatherMap — urban + 4 rural baseline temps
- Open-Meteo — 30-day historical + ET_0 vegetation proxy
- OpenStreetMap Overpass API — real building/road density
- ONNX Runtime — pre-trained UHI prediction model
- Groq AI SDK — natural language insights + chatbot

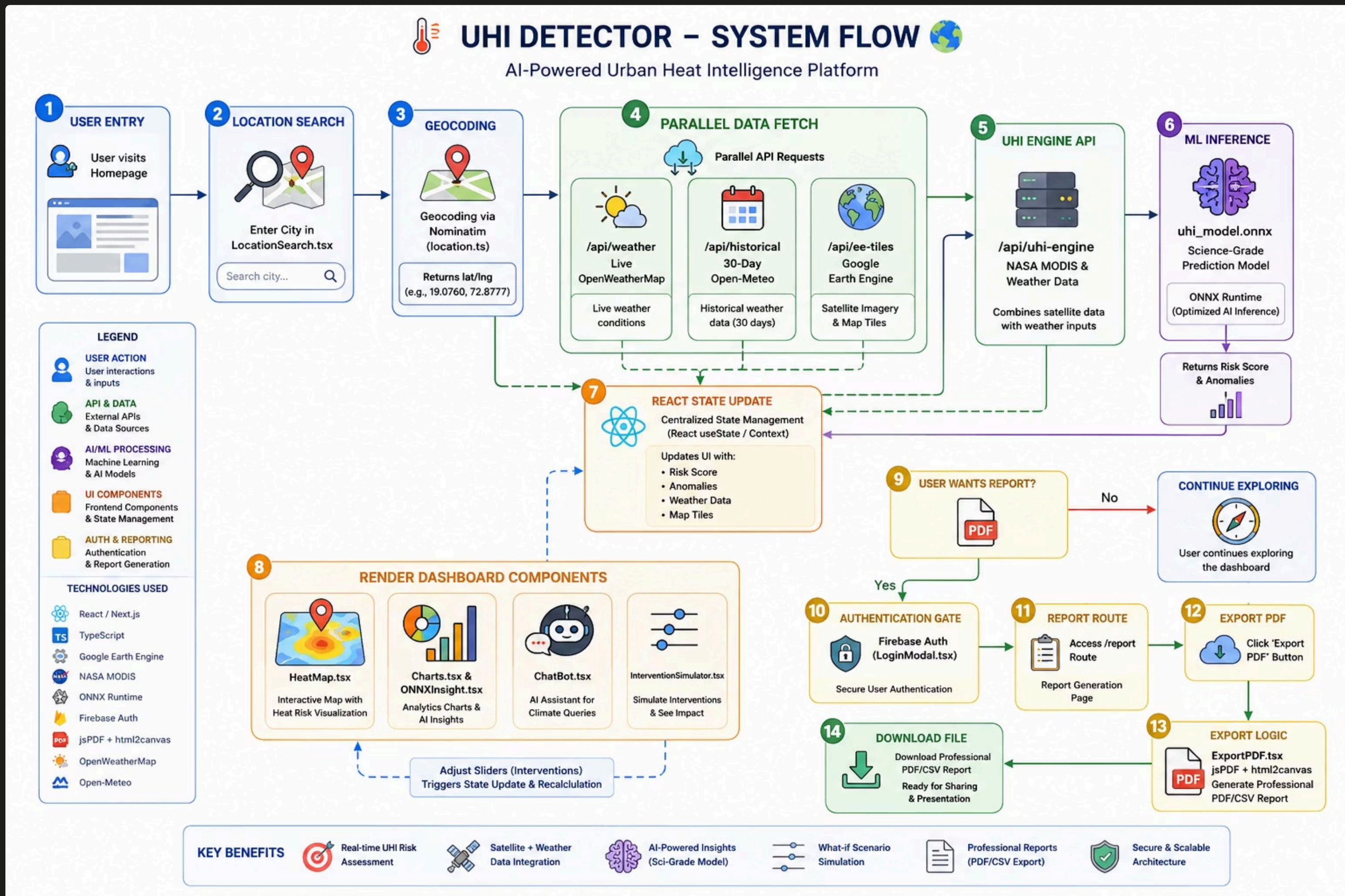
⚡ Product Layer

- Next.js 16 (App Router) + React 19
- Tailwind CSS v4 + Framer Motion
- Leaflet — interactive satellite map overlays
- Chart.js — historical + regression forecast charts
- jsPDF + html2canvas — professional PDF export
- Firebase v12 + NextAuth — auth + user management

Every data source is free-tier accessible. **Zero proprietary data dependency.** Infinitely scalable from day one.

SYSTEM DESIGN

From City Name to Insight



Cities are getting hotter.
Planners need better tools.
We built one.

See It Live

💻 GitHub: **https://github.com/jjf2009/heat_watch**

The Team

Built with 💙 at **PCCE E-Cell Hackathon 2026**

Team RAMpage

HeatWatch — Science-Grade Urban Heat Intelligence